

BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project Report

Submitted By:

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Technology Stack:

Python

Pandas

SQLite

Power BI

Project Duration:

Day 1- Day 7

June 2026

Executive Summary

The Bluestock Mutual Fund Analytics project is an end-to-end data analytics solution developed to analyze mutual fund industry data using Python, SQLite, and Power BI.

The project covers the complete analytics lifecycle including data ingestion, cleaning, transformation, exploratory data analysis, performance analytics, advanced risk analytics, and dashboard development.

More than 10 datasets were processed and transformed into a structured analytics environment. Key metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were calculated.

An interactive Power BI dashboard was developed to help investors and analysts monitor fund performance, investor behavior, SIP trends, sector allocation, and risk measures.

The final solution demonstrates how data-driven insights can support mutual fund investment decisions.

Project Objectives

1. Collect and organize mutual fund datasets.
2. Clean and validate raw data.
3. Build an ETL pipeline.
4. Store processed data in SQLite.
5. Perform Exploratory Data Analysis.
6. Calculate performance metrics.
7. Perform advanced risk analytics.
8. Create interactive Power BI dashboards.
9. Generate investor insights.
10. Deliver a complete analytics solution.

Data Sources

S No	Dataset
1	01_fund_master.csv
2	02_nav_history.csv
3	03_aum_by_fund_house.csv
4	04_monthly_sip_inflows.csv
5	05_category_inflows.csv
6	06_industry_folio_count.csv
7	07_scheme_performance.csv
8	08_investor_transactions.csv
9	09_portfolio_holdings.csv
10	10_benchmark_indices.csv

Total Datasets Processed: 10

ETL Architecture

```
bluestock_mf_capstone/
├── data/
│   ├── raw/      ← original downloaded files
│   ├── processed/ ← cleaned, merged CSVs
│   └── db/        ← bluestock_mf.db (SQLite)
├── notebooks/
│   ├── 01_data_ingestion.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_eda_analysis.ipynb
│   ├── 04_performance_analytics.ipynb
│   └── 05_advanced_analytics.ipynb
├── scripts/
│   ├── etl_pipeline.py
│   ├── live_nav_fetch.py
│   ├── compute_metrics.py
│   └── recommender.py
├── sql/
│   ├── schema.sql
│   └── queries.sql
├── dashboard/
│   └── bluestock_mf.pbix
├── reports/
│   ├── Final_Report.pdf
│   └── Presentation.pptx
└── README.md
```

ETL Architecture

The Mutual Fund Analytics project follows a structured ETL (Extract, Transform, Load) architecture to ensure that raw datasets are converted into clean and analysis-ready data.

The ETL process consists of three major phases:

1. Extract
2. Transform
3. Load

1. Extract

- * Raw CSV files were collected from the project datasets.
- * Ten different datasets were used including fund master data, NAV history, SIP inflows, investor transactions, portfolio holdings, and benchmark indices.

2. Transform

- * Missing values were identified and handled.
- * Duplicate records were removed.
- * Date columns were converted into standard datetime format.
- * Numeric columns were validated and corrected.
- * Invalid values such as negative NAVs and transaction amounts were filtered.

3. Load

- * Cleaned datasets were stored inside the processed folder.
- * Data was loaded into SQLite using SQLAlchemy.
- * Fact and dimension tables were created for analytical queries and dashboard reporting.

The ETL pipeline ensures consistency, accuracy, and reliability of data before performing business analytics.

Data Cleaning and Validation

Data quality plays a critical role in ensuring accurate analytical results. Therefore, extensive cleaning and validation activities were performed on all datasets.

The following validation checks were applied:

- Missing Value Handling

Missing values were identified and replaced using appropriate methods such as forward filling and default values.

- Duplicate Removal

Duplicate records were detected and removed to avoid double counting.

- Date Standardization

Various date formats were converted into a uniform datetime format.

- Numerical Validation

NAV values, transaction amounts, AUM values, and return percentages were validated to ensure they were numeric and positive.

- Business Rule Validation

Expense ratios were verified within acceptable ranges.

Transaction types were standardized into SIP, Lumpsum, and Redemption categories.

The final processed datasets achieved high data quality and were suitable for advanced analytics and dashboard creation.

Database Design

A SQLite database was developed to serve as the analytical storage layer of the project.

The database stores all cleaned datasets and supports efficient querying and dashboard integration.

Database Components:

Dimension Tables:

- dim_fund
- dim_date

Fact Tables:

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

Primary and foreign key relationships were established to maintain data integrity.

Benefits of SQLite:

- Lightweight and easy to deploy
- Fast analytical queries
- Seamless integration with Python and Power BI
- No separate database server required

The database file created for this project is named bluestock_mf.db.

NAV Trend Analysis

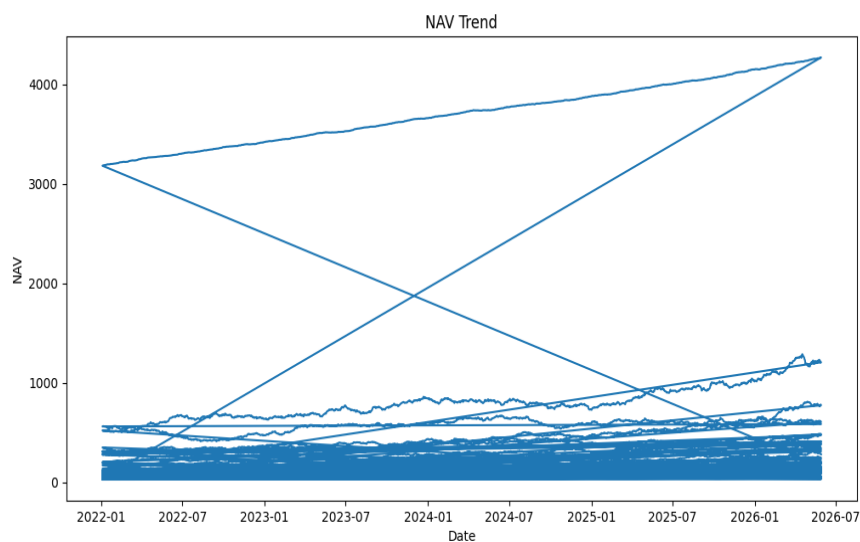
Net Asset Value (NAV) represents the per-unit value of a mutual fund.

Daily NAV trends from 2022 to 2025 were analyzed for multiple schemes.

Observations:

- Most funds exhibited a positive long-term growth trend.
- Strong market performance during 2023 resulted in noticeable NAV appreciation.
- Temporary corrections were observed during market volatility periods.
- Equity-oriented funds demonstrated higher NAV growth compared to debt-oriented funds.

The NAV trend analysis helps investors understand fund growth behavior over time.



nav_trend.png

AUM & SIP Analysis

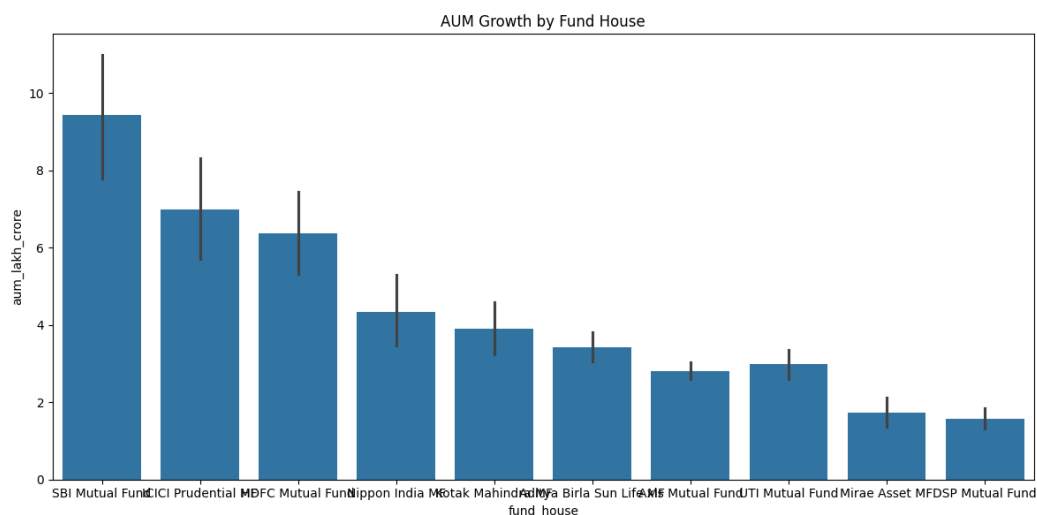
Assets Under Management (AUM) indicates the total market value of assets managed by mutual funds.

Monthly SIP inflow analysis was performed to evaluate investor participation trends.

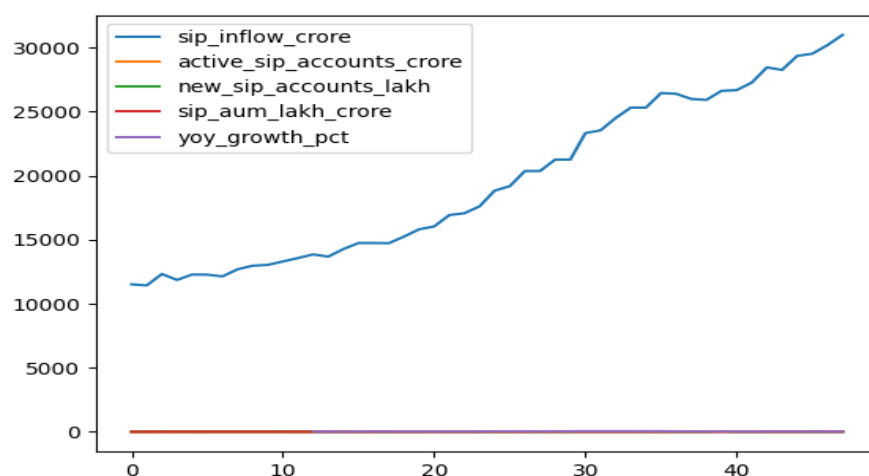
Key Observations:

- Industry AUM increased consistently from 2022 to 2025.
- SBI Mutual Fund maintained market leadership in terms of AUM.
- SIP inflows reached record levels in 2025.
- Retail investor participation increased significantly through systematic investment plans.

The growth in SIP contributions reflects increasing investor confidence in mutual funds as a long-term investment vehicle.



aum_growth.png



sip_trend.png

Category and Sector Analysis

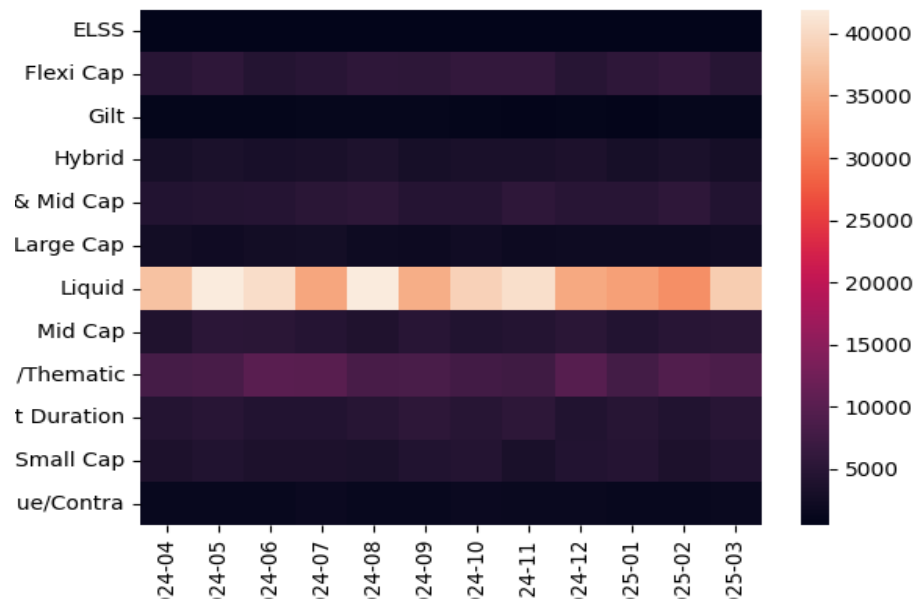
Category inflow analysis was conducted to identify investor preferences across mutual fund categories.

Sector allocation analysis was performed using portfolio holdings data.

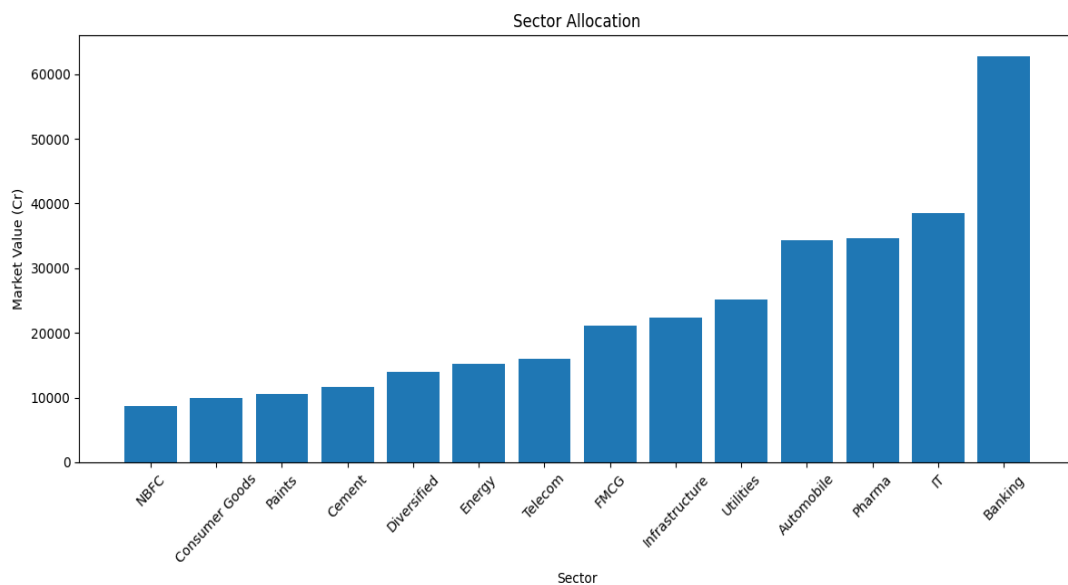
Key Findings:

- Equity categories attracted the highest net inflows.
- Hybrid funds experienced moderate investor interest.
- Debt categories showed relatively stable inflows.
- Financial Services sector held the largest portfolio allocation.
- Technology and Banking sectors also represented significant portfolio weights.

Sector allocation analysis helps investors understand portfolio diversification and concentration risk.



category_heatmap.png



sector_allocation.png

Performance Analytics

Performance metrics were calculated to evaluate the efficiency and effectiveness of mutual funds.

Metrics Calculated:

1. CAGR (Compound Annual Growth Rate)

Measures annualized investment growth.

2. Sharpe Ratio

Measures risk-adjusted return.

3. Sortino Ratio

Measures return relative to downside risk.

4. Alpha

Measures excess return compared to benchmark.

5. Beta

Measures sensitivity to market movements.

6. Maximum Drawdown

Measures the largest observed loss from peak value.

These metrics provide a comprehensive framework for comparing mutual fund performance across different schemes.

Risk Analytics

Risk analytics evaluates the potential downside exposure of mutual fund investments.

Value at Risk (VaR):

Measures the maximum expected loss under normal market conditions with 95% confidence.

Conditional Value at Risk (CVaR):

Measures the average loss beyond the VaR threshold.

Benefits:

- Helps investors understand downside risk.
- Supports portfolio risk management.
- Enables comparison of risk levels across funds.

Funds with higher VaR and CVaR values are generally considered more volatile and risky.

Advanced Analytics

Advanced analytics techniques were implemented to generate deeper business insights.

Rolling Sharpe Ratio:

Tracks risk-adjusted performance over time.

Investor Cohort Analysis:

Groups investors based on their first investment year.

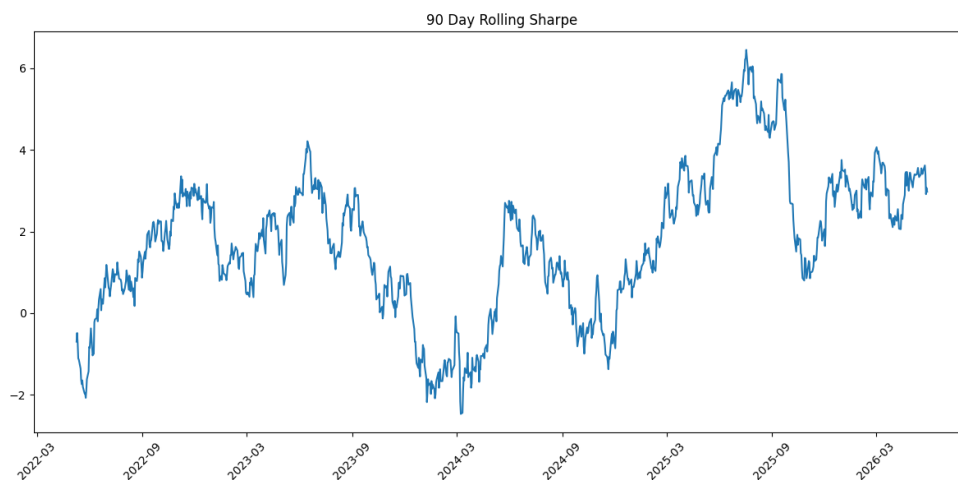
SIP Continuity Analysis:

Measures consistency of SIP investments and identifies at-risk investors.

Herfindahl-Hirschman Index (HHI):

Measures sector concentration within portfolios.

These techniques help identify investment patterns, investor behavior, and portfolio concentration risks.



rolling_sharpe_chart.png

Power BI Dashboard

➤ Power BI Dashboard – Industry Overview

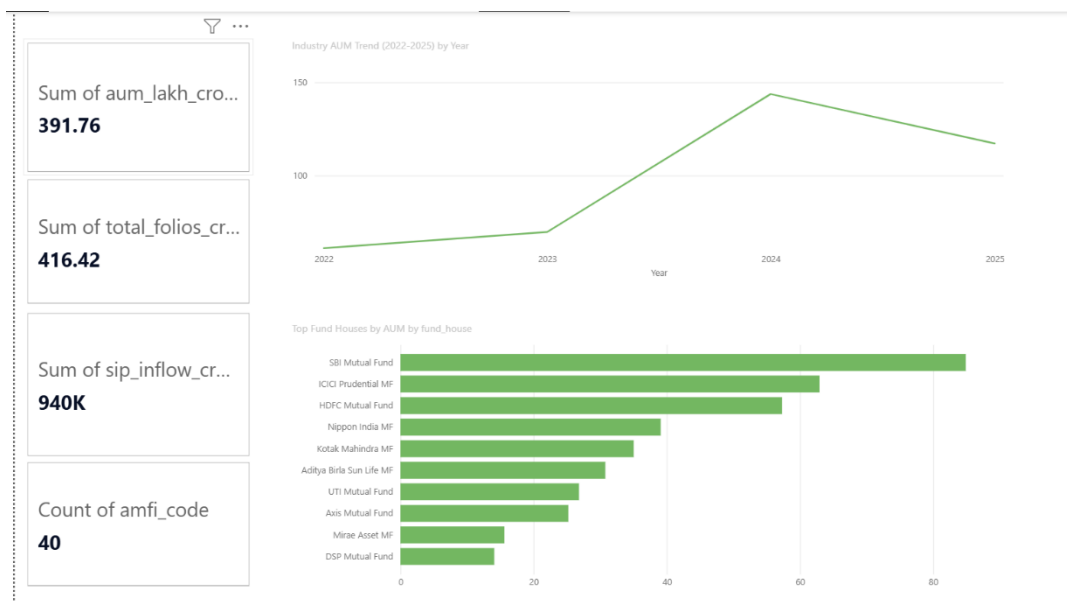
The Industry Overview dashboard provides a high-level summary of the Indian mutual fund industry.

Key Performance Indicators (KPIs) displayed on this page include:

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Folios
- Total Schemes

The dashboard also includes:

- Industry AUM growth trend from 2022 to 2025
- Fund House-wise AUM comparison
- Market growth indicators



industry_overview.png

Insights:

1. The mutual fund industry showed consistent growth in AUM during the analysis period.
2. SIP inflows increased significantly, indicating strong retail investor participation.
3. The number of active folios continued to grow, reflecting increasing adoption of mutual fund investments.
4. . Large Asset Management Companies (AMCs) contributed a major share of total industry AUM.

This dashboard serves as an executive summary page for business users and investors.

➤ Power BI – Fund Performance

The Fund Performance dashboard focuses on evaluating and comparing mutual fund performance.

Visualizations included:

- Risk vs Return Scatter Plot
- Fund Scorecard Table
- NAV vs Benchmark Comparison
- Fund Ranking Analysis

The dashboard enables users to compare schemes using multiple performance indicators such as:

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha

- Beta
- Maximum Drawdown

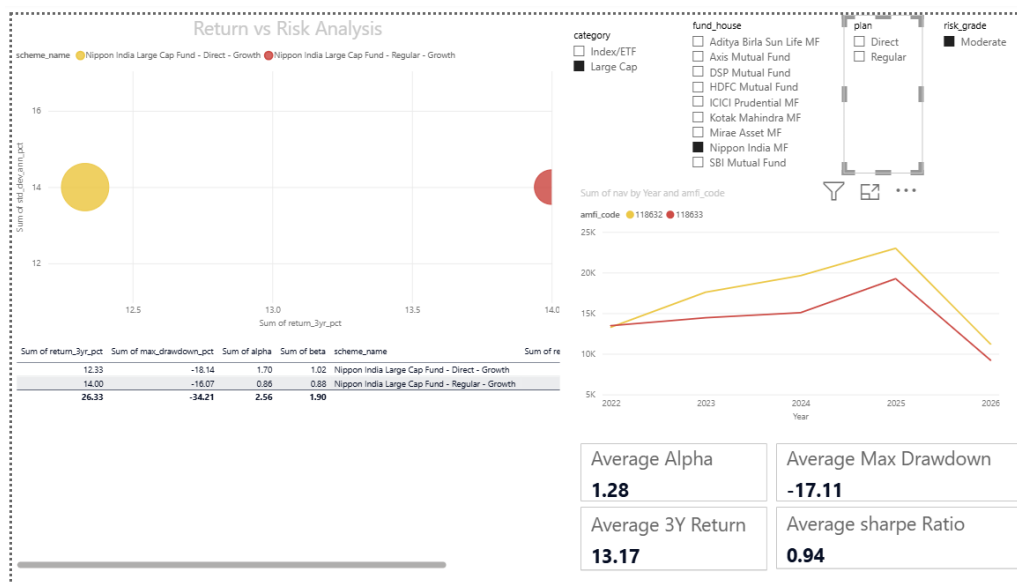
Interactive slicers allow filtering by:

- Fund House
- Category
- Investment Plan

Insights:

1. Funds with higher Sharpe Ratios generated superior risk-adjusted returns.
2. Certain funds consistently outperformed benchmark indices.
3. High-return funds generally exhibited higher volatility.
4. Performance rankings helped identify top-performing schemes within each category.

This dashboard supports informed investment decision-making and fund comparison.



fund_performance.png

➤ Power BI Dashboard: Investor Analytics & SIP Trends

The Investor Analytics dashboard provides insights into investor behavior, transaction patterns, and SIP growth trends.

Visualizations included:

- State-wise Transaction Analysis
- SIP, Lumpsum, and Redemption Distribution
- Age Group Analysis
- Monthly Transaction Trends
- SIP Inflow Trends
- Category Inflow Heatmaps

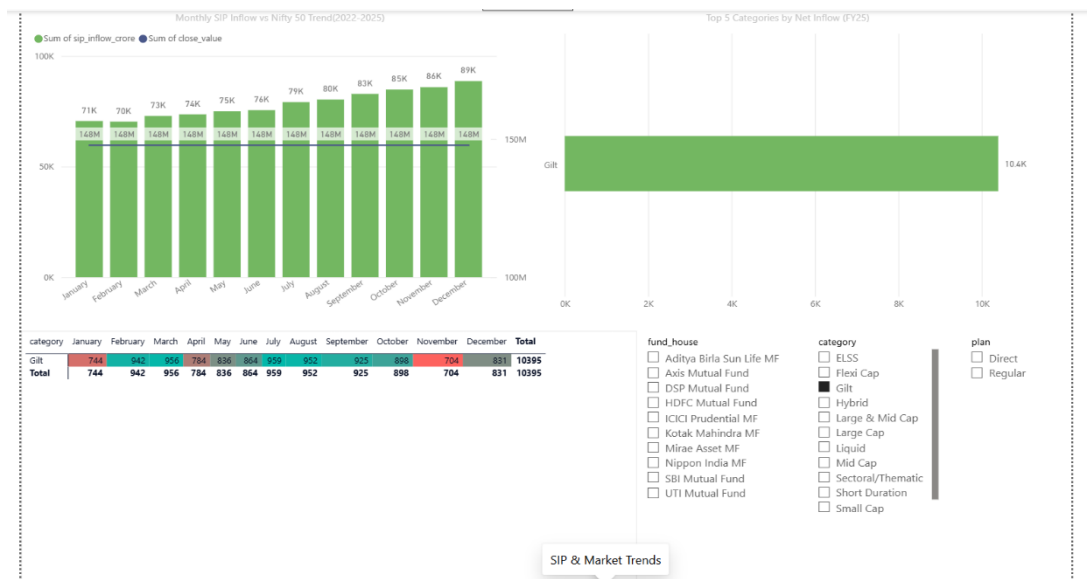
Key Insights:

1. Certain states contributed a larger share of mutual fund investments.
2. SIP transactions represented the majority of recurring investments.
3. Younger investors showed increasing participation in equity-oriented schemes.
4. Monthly SIP inflows demonstrated steady growth throughout the study period.
5. Equity categories attracted the highest net inflows compared to debt and hybrid categories.

The dashboard enables stakeholders to understand investor preferences and monitor market participation trends.



investor_analytics.png



market_trends.png



nav_detail.png

Dashboard Business Value

The Power BI dashboard transforms complex mutual fund data into an interactive and user-friendly reporting solution.

Business Benefits:

- Faster decision-making through real-time visual insights.
- Easy comparison of mutual fund performance.
- Identification of investment trends and risk patterns.
- Improved understanding of investor behavior.
- Enhanced reporting for analysts, fund managers, and investors.

The dashboard integrates ETL outputs, analytical metrics, and business intelligence tools into a single platform, providing a comprehensive view of mutual fund industry performance.